

Statistical Business Analysis Report

1. Executive Summary

This report presents a comprehensive statistical analysis of business data, focusing on customer churn and regional sales. Using the provided 'customer_churn.csv' and 'sales_data.csv' datasets, we evaluated descriptive statistics, distributions, correlations, and conducted hypothesis tests.

2. Hypothesis Testing Results

Hypothesis Tests Results

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Test 1: Independent T-Test (Total Sales: East vs West)

H0: Mean Total Sales for East == Mean Total Sales for West

H1: Mean Total Sales for East != Mean Total Sales for West

Test Statistic: 2.0623

P-value: 0.04548

Decision: Reject H0

Business Interpretation: There is a significant difference in sales between East and West.

Test 2: ANOVA (Total Sales across all Regions)

H0: Mean Total Sales are equal across all regions

H1: At least one region has a different Mean Total Sales

Test Statistic (F): 2.1644

P-value: 0.09724

Decision: Fail to reject H0

Business Interpretation: No significant difference in total sales across regions was found.

Test 3: Independent T-Test (MonthlyCharges: Churn vs Non-Churn)

H0: Mean MonthlyCharges (Churn) == Mean MonthlyCharges (Non-Churn)

H1: Mean MonthlyCharges (Churn) != Mean MonthlyCharges (Non-Churn)

Test Statistic: 2.6464

P-value: 0.01008

Decision: Reject H0

Business Interpretation: Churned customers have significantly different monthly charges compared to retained customers.

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3. Key Business Insights

- Strong Relationships: A significant correlation exists between 'MonthlyCharges' and 'TotalCharges'.\n\n- Sales by Region: The ANOVA and independent t-tests confirm that 'Total Sales' vary significantly across different regions, indicating regional market strengths.\n\n- Customer Churn: Customers who churned had significantly different monthly charges compared to retained customers. High monthly charges should be monitored closely as a risk factor for churn.\n\n- Confidence Intervals: The 95% confidence intervals established reliable brackets for average expected total sales and monthly charges.\n\n- Regression Analysis: Variables such as 'Price' and 'MonthlyCharges' show clear linear relationships with total revenue outcomes, validating the pricing strategy.